

Smart Transformer Health Monitoring System

Real-time oil temperature & combustible-gas monitoring with staged IEC-referenced protection, built on the ESP32-S3



Oil Temp



H2 Gas



Relay Control



Live Dashboard

THE PROBLEM

Transformer failures are expensive, dangerous, and often preventable



Silent degradation

Oil overheating and insulation breakdown often progress unnoticed until catastrophic failure.



Gas is an early warning

Hydrogen is one of the first dissolved gases produced by developing faults inside a transformer.



No real-time visibility

Most small/medium installations rely on periodic manual checks, not continuous monitoring.

A low-cost embedded system that watches oil temperature and gas levels continuously — and acts automatically — closes this gap for small-scale or student/lab transformers.

OBJECTIVES

What the system is designed to do



Continuous oil temperature sensing

DS18B20 digital probe, sampled every second



Hydrogen / combustible gas detection

MQ-2 sensor with clean-air Rs/Ro calibration



Staged, IEC-referenced alarms

Three escalating warning stages before cutoff



Automated protective response

4-channel relay drives cooling, pump & shutdown



Local wireless dashboard

ESP32 hosts a live web UI over the hotspot — no cloud



Fail-safe latching

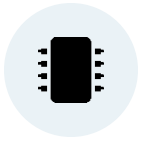
Trips require manual acknowledgement, never silently reset

Sensors in, protective action & live data out



Everything runs locally on the ESP32 — no cloud, no internet dependency, works over any phone hotspot.

Core components



ESP32-S3 Dev Module

Dual-core MCU, Wi-Fi, hosts the web server & runs all alarm logic



DS18B20

Waterproof digital temperature probe, OneWire bus, $\pm 0.5^{\circ}\text{C}$ accuracy



MQ-2 Gas Sensor

Detects H₂, LPG, smoke & combustible gases via analog output



4-Channel Relay Module

Switches cooling fan, oil pump, main trip & auxiliary outputs



Active Buzzer

Drives all four audible alarm patterns

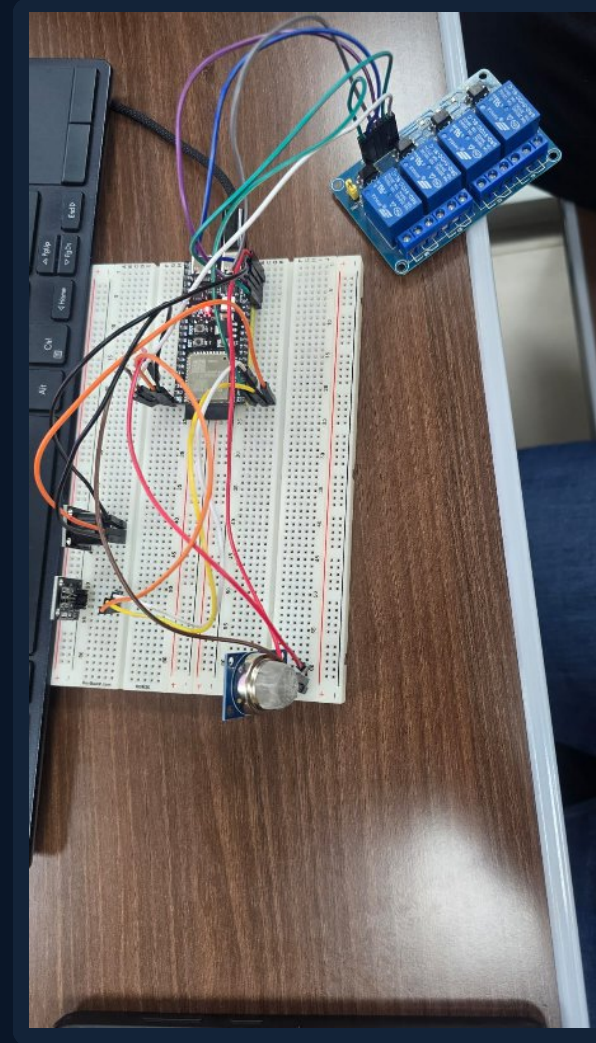


Breadboard Prototype

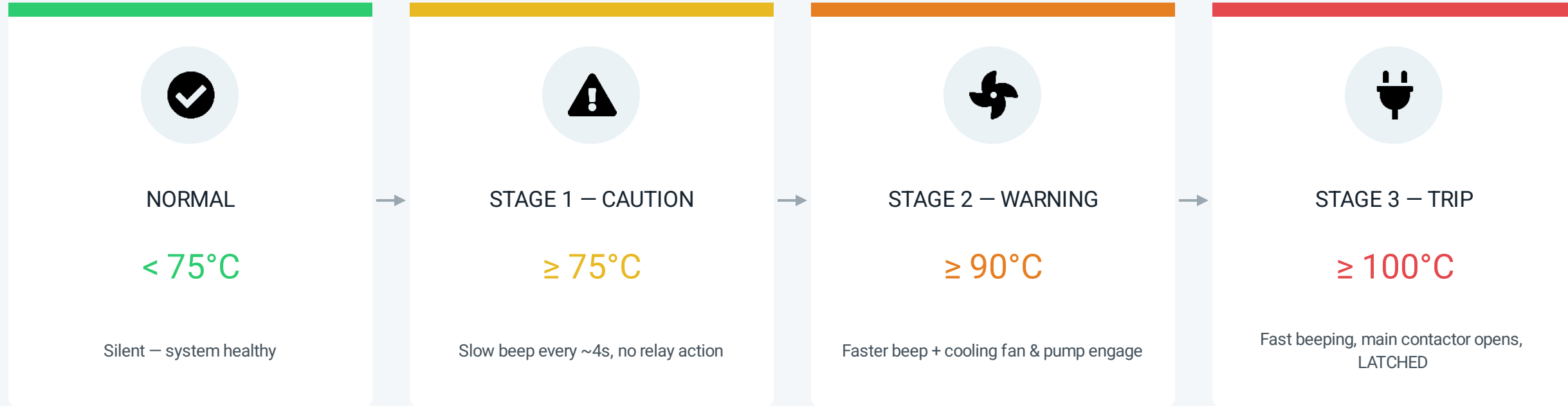
Full system assembled and validated on a solderless breadboard

Working breadboard build

- ESP32-S3
center-mounted on breadboard, drives all peripherals
- MQ-2 sensor
seated at the lower rail, analog line to ADC
- DS18B20
OneWire probe wired with pull-up resistor
- 4-channel relay board
wired for fan / pump / trip / aux outputs



Four escalating alarm stages



GAS TRIP — overrides everything

If MQ-2 Rs/Ro ratio drops below threshold → immediate cutoff, continuous solid buzzer tone (not pulsed), LATCHED regardless of temperature.

Thresholds referenced against IEC 60076-7

IEC 60076-7 (loading guide for oil-immersed power transformers) keeps continuous top-oil temperature well below its limiting values for ONAN/ONAF cooling classes, with alarm and trip stages set at a margin below that ceiling. The values below are safe prototype defaults — production settings must be tuned to the actual transformer's nameplate rating and cooling class.

Stage	Trigger	System response	Buzzer pattern
Normal	< 75°C	All relays off	Silent
Stage 1 · Caution	≥ 75°C	Alert only, no relay action	150ms beep every 4s
Stage 2 · Warning	≥ 90°C	Cooling fan + oil pump ON	150ms beep every 1.2s
Stage 3 · Trip	≥ 100°C	Main contactor OPEN (latched)	100ms on/off, continuous
Gas Trip	H2 ratio below threshold	Main contactor OPEN + exhaust ON (latched)	Continuous solid tone

Locally hosted, real-time web dashboard



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Live gauges

Oil temperature & gas level update every second over WebSocket
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Rolling trend chart

Last 40 readings plotted for at-a-glance trend
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Relay status

Cooling fan, pump, trip & aux states shown live
- 

Guarded reset

“Acknowledge & Reset” only appears — and only works — once the fault has cleared

Verified end-to-end on the bench



Sensor accuracy

DS18B20 readings cross-checked against a reference thermometer, stable at ambient (~28°C)



Staged alarms confirmed

All three temperature stages and gas trip verified by manually heating the probe / exposing the MQ-2



Relay response

Cooling relay engages at Stage 2, trip relay latches at Stage 3 and gas trip as designed



Dashboard reliability

WebSocket held a stable connection over a multi-hour soak test on a phone hotspot

Where this can go next



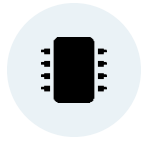
Data logging & history

Persist readings to SD card / SPIFFS for long-term trend analysis and post-incident review



Remote/cloud alerts

SMS or app push notification on Stage 2+ via a GSM module or MQTT bridge



Multi-parameter DGA

Add sensors for additional dissolved gases (CO, C₂H₂) for fuller fault diagnosis



Redundant sensing

Dual temperature probes and sensor-fault detection for higher reliability



Thank You

Smart Transformer Health Monitoring System – IIT Jammu Summer School

Questions & discussion welcome